Epoch Evolution of the MOND Crossover Scale in Density Field Dynamics: An Epoch-Consistency Argument for $a_\star(z) = 2\sqrt{\alpha}\, cH(z)$

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We argue that within Density Field Dynamics (DFD), the MOND crossover acceleration evolves with cosmic epoch as $a_\star(z) = 2\sqrt{\alpha}\, cH(z)$, where α is the fine-structure constant and $H(z)$ is the Hubble rate at redshift z. The derivation uses the two topological invariants $k_a = 3/(8\alpha)$ (from $\mathbb{C}P^2 \times S^3$ gauge emergence) and $q_{S^3} = 3/2$ (the S^3 scaling charge), together with a conditional uniqueness proposition: under those assumptions (locality of the DFD action plus FRW homogeneity plus the absence of any frozen external IR calibration), $cH(t)$ is the unique instantaneous cosmic scale available to the coarse-grained IR functional at cosmic time t. The argument is the epoch-z analog of the $z = 0$ variational-stationarity derivation of $a_\star = 2\sqrt{\alpha}\, cH_0$ ([2], Appendix N), now applied at arbitrary redshift under the same epoch-consistency rule that v3.4 already uses to promote the cosmological closure $G(t)\hbar H(t)^2/c^5 = \alpha^{57}$ from a present-epoch to an all-epoch statement ([2], Section XIX.D). With that promotion accepted, $a_\star(z) \propto cH(z)$ is the companion evolution law. Since atomic units are epoch-invariant under DFD's epoch-invariant fundamental constants $(\alpha, m_e, \hbar)$, the prediction is directly comparable to JWST rotation-curve measurements after standard kinematic, inclination, and mass-inference reductions: the MOND transition scale should appear enhanced by a factor $H(1)/H_0 \approx 1.79$ at $z \sim 1$ relative to the local-universe value. Observation of a frozen $a_\star$ at high redshift, with the empirically inferred background history, would falsify the proposition.

I. INTRODUCTION

The MOND acceleration scale $a_0 \approx 1.2\times10^{-10}\,\mathrm{m/s^2}$ [1] is numerically close to $cH_0 \approx 6.5 \times 10^{-10}\,\mathrm{m/s^2}$, a coincidence that has resisted explanation in most non-ΛCDM frameworks. Within Density Field Dynamics (DFD) [2], this coincidence is resolved: $a_\star = 2\sqrt{\alpha}\, cH_0$ is derived from two topological invariants by variational stationarity of a coarse-grained infrared (IR) functional [2], Appendix N.

The $z = 0$ derivation leaves open a physical question: what does $a_\star$ equal at arbitrary epoch z? Two candidate answers exist in the literature on MOND cosmology:

(i) *Frozen*: $a_\star(z) = 2\sqrt{\alpha}\, cH_0$ for all z, with the IR scale fixed at its present value.

(ii) *Evolving*: $a_\star(z) = 2\sqrt{\alpha}\, cH(z)$, tracking the local cosmic Hubble rate.

The two predictions differ by a factor of $H(z)/H_0 \approx 1.8$ at $z = 1$ (in a ΛCDM-like background) and by larger factors at higher z. JWST rotation-curve surveys of intermediate-redshift galaxies are now reaching the precision where this distinction becomes observationally accessible.

In this paper we argue that within DFD, answer (ii) is the consistent extension of v3.4's structure. The argument uses three inputs:

(A) The topological coefficient $k_a = 3/(8\alpha)$, from $\mathbb{C}P^2 \times S^3$ gauge emergence ([2], Appendix G).

(B) The S^3 scaling charge $q_{S^3} = 3/2$, from the Chern-Simons partition function on S^3 ([2], Appendix N).

(C) A conditional-uniqueness proposition for the cosmic IR scale: under locality of the DFD action plus FRW homogeneity plus the absence of a frozen external IR calibration and the absence of any additional cosmic scalar beyond $H(t)$, the unique inverse-time scale available to the coarse-grained IR functional at cosmic time t is $cH(t)$.

Inputs (A) and (B) are epoch-invariant topological quantities. Input (C) is a proposition whose hypotheses are discussed explicitly in Section III. We also argue that accepting the epoch-invariant promotion of the DFD cosmological closure from its present-epoch form $G\hbar H_0^2/c^5 = \alpha^{57}$ to its all-epoch form $G(t)\hbar H(t)^2/c^5 = \alpha^{57}$ ([2], Section XIX.D, Eqs. 516-519) forces, by the same epoch-consistency rule, use of $cH(t)$ as the IR scale in the Appendix N functional at cosmic time t. With that promotion accepted, the extension of $a_\star$ to $a_\star(z) = 2\sqrt{\alpha}\, cH(z)$ is a direct corollary rather than a sidecar argument.

Section II summarizes the $z = 0$ derivation. Section III states the conditional-uniqueness proposition for the cosmic IR scale and derives it two ways: from locality under stated hypotheses, and from consistency with the Section XIX.D closure promotion. Section IV states the main result. Section V works out the companion role of $G(t)$ evolution and clarifies what JWST observes. Section VI gives the JWST falsification test. Section VII discusses the status of the result relative to prior DFD publications and lists hypotheses under which it could fail.

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II. PRELIMINARIES: THE $z = 0$ DERIVATION

A. The local dimensionless invariant

Given DFD postulates (flat $\mathbb{R}^3$, scalar field ψ, matter acceleration $\mathbf{a} = (c^2/2)\nabla\psi$) and a global μ-closure, the onset of non-Newtonian response can depend only on the dimensionless scalar built from the local acceleration $|\mathbf{a}|$ and a cosmic IR scale. At the present epoch, this scale is cH_0, and the invariant is

$$\Xi(\mathbf{x}) := k_a \left(\frac{|\mathbf{a}(\mathbf{x})|}{cH_0} \right)^2, \tag{1}$$

where $k_a = 3/(8\alpha)$.

B. Scaling charge and IR functional

Lemma 1 ([2], Lemma N.12)**.** *For $SU(2)$ Chern-Simons theory on S^3, the partition function satisfies* $\log Z_{S^3}(k) = \text{const} - \frac{3}{2}\log(k+2) + O(k^{-2})$. *The dimensionless scaling charge is* $q_{S^3} := -\partial \log Z_{S^3}/\partial \log(k+2) = 3/2$.

The minimal coarse-grained IR functional built from Ξ and q_{S^3} is

$$\mathcal{S}[\psi] := \int_\Omega d^3x \left(\Xi(\mathbf{x}) - q_{S^3} \log \Xi(\mathbf{x}) \right). \tag{2}$$

C. Stationarity and $a_\star$

Theorem 2 ([2], Theorem N.13)**.** *For the scaled family $\psi_\lambda := \lambda\psi_0$, the functional $\mathcal{S}$ is stationary with respect to λ at $\lambda_*^2 = q_{S^3}/\overline{\Xi}_0$, where $\overline{\Xi}_0 := V^{-1}\int_\Omega \Xi_0\, d^3x$. The mean crossover invariant is*

$$\overline{\Xi}_* = q_{S^3} = \frac{3}{2}. \tag{3}$$

Corollary 3 ([2], Theorem N.14)**.** *In the homogeneous-gradient limit, $\overline{\Xi}_* = \Xi_*$ and*

$$a_\star = cH_0\sqrt{\frac{\Xi_*}{k_a}} = cH_0\sqrt{\frac{3/2}{3/(8\alpha)}} = 2\sqrt{\alpha}\, cH_0. \tag{4}$$

III. CONDITIONAL UNIQUENESS OF THE COSMIC IR SCALE

Proposition 4 (Conditional uniqueness)**.** *Consider the coarse-grained IR functional describing local DFD dynamics at cosmic time t_z in the rest frame of a comoving observer. Assume:*

(H1) Locality. *The DFD action is local on flat $\mathbb{R}^3 \times \mathbb{R}$ ([2], Appendix U), so the instantaneous field equation at (x,t) depends only on $\rho(x,t)$, $\psi(x,t)$, and spatial derivatives at time t.*

(H2) FRW homogeneity. *The cosmic background is well described by an FRW metric with single scalar Hubble rate $H(t)$; no other cosmic scalar with dimensions of inverse time is available.*

(H3) No frozen external calibration. *The coefficient $a_\star$ of the crossover function $\mu(|\nabla\psi|/a_\star)$ is not treated as a constant of nature fixed by past microsector matching and then inherited; it is instead determined afresh by the instantaneous coarse-grained functional at each epoch.*

(H4) No additional cosmic scalar. *No additional dimensional quantity enters the functional beyond c, α, topological numbers, and $H(t)$.*

Then $a_\star(t) = (\text{topological number}) \times cH(t)$, with the topological coefficient fixed by variational stationarity.

Proof. By (H1) the instantaneous field equation depends only on time-t data. By (H2) the only inverse-time scalar available from cosmic data at time t is $H(t)$. By (H4) no additional scalar enters. By (H3) the coefficient $a_\star$ is not a frozen constant but determined at each epoch. Dimensional analysis then leaves $cH(t)$ as the unique inverse-length scale times speed of light available for $a_\star$, up to a topological prefactor. The prefactor is determined by the stationarity argument of Section IV. □

We emphasize what the proposition does *not* say. Locality alone is compatible with $a_\star$ being a microsector-fixed constant inherited by the EFT; that scenario is excluded by (H3). The proposition is therefore conditional on (H3)'s rejection of the frozen-calibration possibility. We justify that rejection below by a consistency argument tied to v3.4's Section XIX.D.

A. Consistency argument via Section XIX.D closure promotion

In v3.4 the DFD cosmological closure is first stated at the present epoch as $G\hbar H_0^2/c^5 = \alpha^{57}$ ([2], Appendix O, Theorem O.3). Section XIX.D then promotes it to an all-epoch statement $G(t)\hbar H(t)^2/c^5 = \alpha^{57}$ ([2], Eqs. (516)-(519)), from which the photon-frame $G(t) \propto 1/H(t)^2$ evolution follows.

This promotion already adopts the rule "evaluate each cosmic quantity at the epoch of the system." The identity is first derived using present-epoch values of G and H_0; its extension to all epochs uses the same relation with $H(t)$ substituted for H_0. No additional physics input is invoked for the promotion; only consistency of the topological closure at arbitrary cosmic time.

Proposition 4, and the extension $a_\star(z) = 2\sqrt{\alpha}\, cH(z)$ that follows from it, is precisely the same promotion rule applied to the Appendix N IR functional. Accepting the Section XIX.D promotion while denying it for Appendix N would require an additional physical principle distinguishing the two cases. We are not aware of one. Under the assumption that the same epoch-consistency

rule governs both pieces of v3.4, (H3) is imposed and the $a_\star(z)$ extension follows as a direct corollary.

This is the cleanest version of the argument: it does not rest on locality in isolation, but on a consistency bridge between two pieces of v3.4 that both need the same epoch rule to be internally coherent.

IV. MAIN RESULT

Theorem 5 (Epoch evolution of $a_\star$ under Proposition 4)**.** *Assume the hypotheses of Proposition 4, DFD postulates, and the topological values $k_a = 3/(8\alpha)$ and $q_{S^3} = 3/2$. Then the MOND crossover acceleration at cosmic epoch z is*

$$\boxed{a_\star(z) = 2\sqrt{\alpha}\, cH(z)}. \tag{5}$$

Proof. By Proposition 4, the local dimensionless invariant at epoch z is

$$\Xi_z(\mathbf{x}) := k_a \left(\frac{|\mathbf{a}(\mathbf{x})|}{cH(z)}\right)^2. \tag{6}$$

The coefficient $k_a = 3/(8\alpha)$ is an epoch-invariant topological quantity derived from $\mathbb{C}P^2 \times S^3$ gauge emergence ([2], Theorem G.1), and it is independent of cosmic epoch.

The coarse-grained IR functional at epoch z is constructed from Ξ_z and the epoch-invariant scaling charge $q_{S^3} = 3/2$ (Lemma 1):

$$\mathcal{S}_z[\psi] := \int_\Omega d^3x \left(\Xi_z(\mathbf{x}) - q_{S^3} \log \Xi_z(\mathbf{x})\right). \tag{7}$$

The structure of $\mathcal{S}_z$ is identical to $\mathcal{S}$ (Eq. 2); only the IR scale has changed from cH_0 to $cH(z)$.

Apply the scaled family $\psi_\lambda := \lambda\psi_0$. Then $\Xi_{z,\lambda} = \lambda^2 \Xi_{z,0}$, and

$$\mathcal{S}_z[\psi_\lambda] = \lambda^2 V \overline{\Xi}_{z,0} - q_{S^3}\left(2V \log\lambda + \int_\Omega \log \Xi_{z,0}\, d^3x\right). \tag{8}$$

Differentiating with respect to λ and setting to zero:

$$\frac{d\mathcal{S}_z}{d\lambda} = 2\lambda V \overline{\Xi}_{z,0} - \frac{2q_{S^3}V}{\lambda} = 0, \tag{9}$$

which yields $\lambda_*^2 = q_{S^3}/\overline{\Xi}_{z,0}$. The mean crossover invariant at epoch z is

$$\overline{\Xi}_{z,*} = \lambda_*^2 \overline{\Xi}_{z,0} = q_{S^3} = \frac{3}{2}. \tag{10}$$

In the homogeneous-gradient limit, $\overline{\Xi}_{z,*} = \Xi_{z,*}$, and from Eq. (6):

$$\begin{aligned} a_\star(z)^2 &= \frac{\Xi_{z,*}(cH(z))^2}{k_a} = \frac{(3/2)(cH(z))^2}{3/(8\alpha)} \\ &= 4\alpha(cH(z))^2, \end{aligned} \tag{11}$$

so $a_\star(z) = 2\sqrt{\alpha}\, cH(z)$. □

Corollary 6 (Epoch invariance of Ξ_*)**.** *The dimensionless crossover invariant $\Xi_*(z) = 3/2$ is independent of z. All epoch dependence of $a_\star$ is carried by the IR scale $cH(z)$.*

Corollary 7 (No new topological input)**.** *The proof uses the same two topological quantities (k_a, q_{S^3}) as the $z = 0$ derivation. No additional topological input is required for the epoch extension.*

V. CONSISTENCY WITH THE DFD COSMOLOGICAL CLOSURE

The DFD cosmological invariant ([2], Theorem O.3) is

$$\frac{G(t)\hbar H(t)^2}{c^5} = \alpha^{57}. \tag{12}$$

If the microsector closure holds at all epochs [2], Eq. (516), then the photon-frame gravitational constant evolves as

$$G_{\text{photon}}(t) = \frac{\alpha^{57}c^5}{\hbar H(t)^2}. \tag{13}$$

The companion evolution $a_\star(z) \propto cH(z)$ is its natural partner: Eq. (5) and Eq. (12) are both consequences of the same epoch-invariant topological structure applied to the local cosmic IR scale at each epoch.

A. What JWST actually measures

Fundamental constants α, m_e, $\hbar$ in DFD are epoch-invariant. Therefore the atomic length and time units set by these constants (Bohr radius, Hartree period) are themselves epoch-invariant: a terrestrial meter and second are the same units as a meter and second measured in the local rest frame of a galaxy at $z = 1$.

JWST rotation-curve measurements recover stellar velocities in the galaxy's own local rest frame after removing cosmological redshift and standard reductions (kinematic modeling, inclination correction, beam smearing, baryonic-mass inference), expressed in atomic-frame units. Under Theorem 5, the MOND crossover in the galaxy's local frame occurs at

$$a_\star^{\text{local}}(z) = 2\sqrt{\alpha}\, cH(z) \tag{14}$$

in invariant atomic units. Because atomic units are epoch-invariant, no frame-conversion factor is required; the prediction applies to the measured radial acceleration relation (RAR) transition scale at each redshift, after the standard pipeline reductions above.

The photon-frame G evolution (Eq. 13) is a separate prediction, detectable only through cavity-atom LPI tests ([2], Section XII) where the atomic-photon frame distinction becomes operational. For rotation-curve tests, the atomic-frame prediction Eq. (14) is the clean and direct statement.

VI. FALSIFIABLE PREDICTION

A. The observable at high z

JWST rotation-curve surveys measure velocity $v(r)$ profiles of intermediate-redshift galaxies. The MOND transition radius $r_t(z)$ is the radius at which $a_N(r_t) = a_\star(z)$. In atomic-frame units:

$$r_t(z) = \sqrt{\frac{GM_{\rm bar}}{a_\star^{\rm atomic}(z)}}. \tag{15}$$

For a galaxy of fixed baryonic mass $M_{\rm bar}$, a larger $a_\star$ gives a smaller transition radius, hence the flat-rotation-curve regime sets in at smaller r.

B. Quantitative prediction

Using the empirically inferred background history $H(z)$ with $\Omega_m \approx 0.315$ (consistent with both the ΛCDM phenomenological fit and the DFD ψ-screen interpretation of it; [2], Section XIX.D). The $H(z)/H_0$ ratios below are an observational forecast using the best-fit background, not a derivation from ΛCDM ontology:

TABLE I. DFD predictions for $a_\star$ evolution.

z	$H(z)/H_0$	$a_\star(z)$ $[10^{-10}\,\mathrm{m/s^2}]$	$a_\star(z)/a_\star(0)$
0.0	1.00	1.20	1.00
0.5	1.32	1.58	1.32
1.0	1.79	2.14	1.79
1.5	2.37	2.83	2.37
2.0	3.03	3.63	3.03
3.0	4.57	5.46	4.57

C. Falsification criteria

The theorem is falsified by any of the following:

1. **Frozen $a_\star$**: If JWST rotation curves at $z \sim 1$ show a MOND transition at $a_\star(z=1) = a_\star(z=0) = 1.2\times10^{-10}\,\mathrm{m/s^2}$ to better than 10% precision, Theorem 5 is falsified and the frozen-$a_\star$ alternative (i) is confirmed.
2. **Wrong scaling**: If $a_\star(z)$ evolves but tracks $H(z)^n$ with $n \neq 1$ at $> 3\sigma$ significance (e.g., $n = 1/2$ or $n = 2$), the theorem is falsified.
3. **RAR breakdown**: If galaxies at $z > 1$ fail to show a unified radial acceleration relation (RAR) with scatter comparable to local-universe SPARC data, the coarse-grained IR functional assumption of the proof is invalidated.

D. Distinguishing from ΛCDM

ΛCDM has no intrinsic acceleration scale for galaxy dynamics; the transition between baryon-dominated and dark-matter-dominated rotation is set by the baryonic-to-halo-mass ratio, which varies stochastically between galaxies. ΛCDM does not predict a coherent $a_\star(z)$ evolution tracking $H(z)$.

A systematic trend of transition acceleration $\propto H(z)$ across JWST galaxies would be evidence for DFD's epoch-evolving $a_\star$ prediction and against the stochastic ΛCDM expectation.

VII. DISCUSSION

A. Status relative to prior DFD literature

The scaling $a_\star \propto cH$ has been referenced qualitatively in DFD presentations as a consequence of the α^{57} closure ([2], Section XIX.D, Eq. (516)), but it has not previously been derived explicitly at arbitrary epoch. The present argument closes this gap conditionally: under the hypotheses of Proposition 4 (or equivalently, under the same epoch-consistency rule already used to promote the closure from present-epoch to all-epoch form), Theorem 5 follows as a direct corollary of the variational stationarity derivation applied at arbitrary epoch, using only the same two topological inputs that fix $a_\star$ at $z = 0$.

B. Relation to the Sciama Machian program

Sciama's 1953 proposal [3] that inertia arises from the cosmic matter distribution predicted that local inertial scales should track the cosmic gravitational radius. The DFD relation $a_\star(z) = 2\sqrt{\alpha}\,cH(z)$ is a concrete realization of this expectation: the local MOND transition scale is dynamically locked to the cosmic Hubble rate at each epoch, with the coefficient of proportionality fixed by microsector topology rather than being a free parameter.

This places DFD squarely within the Bondi-Samuel Mach-1 through Mach-6 taxonomy, with Mach-1 ("local inertial frames track cosmic matter distribution") realized through the dynamical coupling $a_\star \leftrightarrow cH(z)$ rather than through a geometric boundary condition.

C. What the theorem does not claim

Several natural extensions are beyond the scope of this theorem:

1. The derivation assumes a coarse-grained IR regime; galaxy-scale dynamics well below the horizon scale are treated as governed by the functional $\mathcal{S}_z$. Cosmological-scale dynamics ($r \sim 1/H$) require separate analysis.

2. The derivation uses the homogeneous-gradient limit (Corollary 3); inhomogeneous cosmic backgrounds may produce corrections at $O(\delta\overline{\Xi}/\overline{\Xi})$.
3. Proposition 4 assumes FRW homogeneity when identifying $H(t)$ as the unique cosmic scalar at time t. Departures from homogeneity at large scales would add corrections. It also assumes no frozen external IR calibration (H3) and no additional cosmic scalar (H4); either assumption failing would invalidate the epoch extension.

D. Experimental priority

JWST rotation-curve measurements at $z \sim 0.5$–2 are the proximal test. Current preliminary data on $z \sim 1$ galaxies (e.g., from JWST/NIRSpec integral-field observations) are approaching the precision required to distinguish frozen-$a_\star$ from evolving-$a_\star$ at $> 3\sigma$. Dedicated surveys optimized for rotation-curve extraction (rather than opportunistic spectroscopy) would resolve the test at 5σ within ~ 3–5 years.

VIII. SUMMARY

Under the DFD postulates and the conditional-uniqueness Proposition 4 (equivalently, under the epoch-consistency rule already used in v3.4 Section XIX.D to promote $G\hbar H^2/c^5 = \alpha^{57}$ from present-epoch to all-epoch form), the MOND crossover acceleration at cosmic epoch z satisfies $a_\star(z) = 2\sqrt{\alpha}\, cH(z)$. The argument extends the $z = 0$ variational-stationarity derivation of [2] (Theorem N.14) to arbitrary epoch, using only the epoch-invariant topological coefficients $k_a = 3/(8\alpha)$ and $q_{S^3} = 3/2$. The crossover dimensionless invariant $\Xi_*(z) = 3/2$ is epoch-invariant; all epoch dependence is carried by the local cosmic IR scale $cH(z)$.

The result is conditional on the hypotheses of Proposition 4: in particular, hypothesis (H3) that $a_\star$ is not frozen by microsector matching at a past epoch. Readers who reject (H3) or who object to the Section XIX.D epoch-consistency bridge can consistently hold the frozen-H_0 alternative instead. The observational test separating the two is the same in either case: JWST-era rotation curves at $z \sim 1$.

The prediction is directly falsifiable: JWST rotation-curve observations at $z \sim 1$ should show an $a_\star$ enhanced by a factor $H(1)/H_0 \approx 1.79$ relative to the present-epoch value. Observation of a frozen $a_\star$ at high z would falsify the result; observation of evolution tracking $H(z)$ would support it.

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